and writing a bit forces us to map the index of the bit to be accessed into a couple of integers.

**Listing 4.4 findBitCoordinates**

```
function
findBitCoordinates(index)
       #1
  byteIndex ← ceil(index / BITS_PER_INT)
                                        #2
  bitOffset ← index mod
BITS_PER_INT                                          #3
  return (byteIndex,
bitOffset)                                               #4
```

#1 Function `findBitCoordinates` is a utility method that given the index of a bit in a bit array, returns the index of the array and the offset of the bit with respect to the array's element at that index.

#2 Given the index of the bit to retrieve, extract the byte index, i.e. which element of the array buffer contains the bit to extract. `BITS_PER_INT` is a (system) constant, whose value is the number of bits used to store an int in the programming language used (for most languages it's 32).

#3 Extract the bit offset inside the buffer byte, in other words the local index of the bit, by performing a modulo operation: we need the rest of the division performed at line #2.

#4 Returns the byte index and bit offset as a couple of values. Note that some programming languages allow